CHAPTER

14

Silver Amalgam

Silver amalgam, the most commonly used restorative material, has withstood all challenges of time. The advantages of silver amalgam always surpass the disadvantages, although problem of mercury has compelled the clinicians to minimize its usage.

Silver amalgam, if properly manipulated, produces a restoration, which could provide many years of service. The failures in amalgam restorations are due to faulty cavity preparation and faulty manipulation. Secondary caries and fracture, the two major failure signs, are attributed to poor manipulation and improper cavity design.

COMPOSITION OF SILVER ALLOY

Amalgam is a mixture of mercury, with one or more metals. An alloy is a union of two or more metals (usually by means of heat). The advantage of alloying metals is that when a new compound is formed, it possesses properties that are not present in single element. Mercury, having the property of dissolving other metals, produces a plastic mass, which has low fusion temperature.

Conventional Silver Alloy

Conventional silver alloy (ADA specification no. 1) contains the following:

Silver (Ag)	68–72% (wt %)
Tin (Sn)	25–27%
Copper (Cu)	2–6%
Zinc (Zn)	0–3%

However, alloys with copper contents more than 6% and may be up to 30% have been introduced with the idea of eliminating γ_2 phase (phase responsible for corrosion) (Fig. 14.1).

An alloy, which contains more than 6% copper is referred to as a high copper alloy.

The early alloys were admixed alloy having silver–copper eutectic and conventional alloy; however, they were modified to make it single composition alloy.



Fig. 14.1: Commercial preparation of low copper (DPI Alloy) and high copper (FusionAlloy) silver alloy

Admixed/Blended Alloy

Admixed alloys are of two types:

i. *Type I*: It contains two parts of conventional lathe cut particles and one part by weight of silver–copper eutectic alloy.

Composition	
Silver	69%
Copper	13%
Tin	17%
Zinc	1%

ii. *Type II*: The reverse of type I alloy, contains two parts of spherical silver-copper alloy and one part of conventional alloy.

Composition	
Silver	60%
Tin	25%
Copper	15%

Single Composition/All in One Alloy

Each particle of the alloy has the same chemical composition; that is why, they are called single composition alloys.

These are of three types:

i. *Ternary alloys*: The composition can be either silver 60%, tin 25%, copper 15% or silver 40%, tin 30%, copper 30%.

ii. *Quaternary alloys*: Silver 60%, copper 13%, tin 24%, indium 3%.
iii. An alloy similar to (i), but containing particles in spheroidal form (i.e. particles are not perfectly spherical).

EFFECT OF CONSTITUENT METALS

Silver

- *Increases strength:* The strength is basically because of unreacted AgSn (γ phase) particles in the matrix. Higher the silver, higher will be the γ-phase.
- *Decreases flow.* The flow is also because of the unreacted AgSn (γ phase) particles. Higher the unreacted particles, lesser will be the flow.
- *Increases setting expansion:* This is because of the formation of γ_1 phase, which is AgHg. The volume of γ_1 phase is greater than both Ag and Hg independently.
- *Reduces setting time:* By accelerating the setting process, since the reaction of silver and mercury is very fast.
- Resists tarnish and corrosion.
- Whitens the alloy.

Tin

- *Reduces strength:* With increase of tin, the AgHg is reduced and SnHg is increased, which is a weaker phase.
- *Helps in amalgamation*: Since tin has great affinity for mercury; controls the fast reaction between silver and mercury because of this affinity.
- *Decreases setting expansion:* The SnHg phase causes the contraction.
- Increases setting time.
- Reduces the resistance to tarnish and corrosion.

Copper

- *Increases strength:* The increased strength is the same as of silver. Only AgSn alloys are brittle and do not show uniform coherence; that is why, some amount of copper is a must.
- *Increases setting expansion:* If copper is more than 5.0% excessive expansion occurs.
- Reduces flow.
- Copper makes the alloy less susceptible to slight imperfections during manipulation by the operator.

Zinc

- Prevents oxidation during manufacturing of alloy.
- Contributes to the workability of silver amalgam during trituration and condensation.
- Responsible for delayed expansion, if zinc-containing alloys are contaminated with water. The expansion occurs after 3–4 days, because zinc reacts with water releasing hydrogen. This hydrogen causes expansion.
- Like copper, it also makes the alloy less susceptible to imperfection during manipulation by the operator.

ALLOY MERCURY REACTION

a. Conventional Alloy

The reaction of alloy and mercury, traditionally take silver and tin, without taking copper and zinc. The reaction is:

$$\underset{\gamma}{AgSn} + Hg \longrightarrow \underset{\gamma_1}{AgHg} + \underset{\gamma_2}{SnHg} + \underset{\gamma}{AgSn}$$

The γ or the AgSn are the unreacted particles. The γ_1 phase is the first phase formed during setting because silver dissolves more quickly in mercury than tin. However, mass transport of mercury into alloy particles might contribute to the formation of new phases equally. After the reaction, the unreacted particles (AgSn) are embedded in a matrix of reaction products with mercury. In conventional amalgams, both γ_1 and γ_2 phases form a continuous network. The distribution of γ_2 phase is important, as it is the phase, which is responsible for corrosion of silver amalgam.

In case of aging amalgam, another phase AgHgSn, i.e. β phase is formed. It has been shown that amalgams remain in a state of non-equilibrium, which might be for one year or even more. The percentage of phases in the conventional alloy is AgHg 66.25%, SnHg 6.25%, AgSn 25% and residual Hg 2.5% (Fig. 14.2).

b. High Copper Alloy

In case of high copper alloys, the reaction occurs in two phases. The first phase is equivalent to the one as shown in conventional alloys, i.e.

$$\underset{\gamma}{AgSn} + Hg \longrightarrow \underset{\gamma_1}{AgHg} + \underset{\gamma_2}{SnHg} + \underset{\gamma}{AgSn}$$

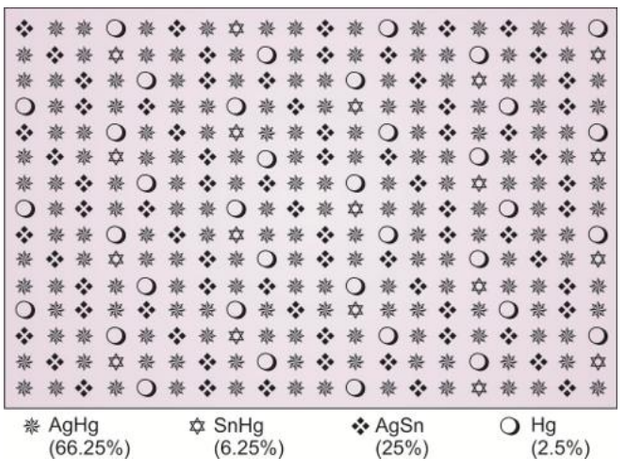


Fig. 14.2: Distribution of phases in conventional alloy

The second phase is the eutectic of silver–copper phase, which are called α_1 and α_2. This type of reaction occurs in admixed type of high copper alloy. In eutectic, two metal phases are present, which are separated but closely adjacent to each other. Here α_1 is silver rich and α_2 is copper rich. These two along with γ (AgSn) react with mercury and again γ_1 (AgHg) and γ_2 (SnHg) are formed. All the tin does not react with mercury; part of tin reacts with the eutectic α_1 and α_2.

$$\underset{\gamma}{AgSn} + \underset{\alpha_1+\alpha_2}{AgCu} + Hg \xrightarrow[\text{mixing}]{\text{After}} \underset{\gamma_1}{AgHg} + \underset{\gamma_2}{Sn_7Hg} + \underset{\gamma}{AgSn} + \ldots + \underset{\varepsilon}{Cu_3Sn} + \underset{\eta}{Cu_6Sn_5} + \underset{\alpha_1+\alpha_2}{AgCu}$$

The reaction goes on. After one week, the SnHg (γ_2) phase reacts completely with the eutectic. The silver forms γ_1 and the tin forms η and ε.

$$\underset{\gamma}{AgSn} + \underset{\alpha_1+\alpha_2}{AgCu} + Hg \xrightarrow[\text{one week}]{\text{After}} \underset{\gamma_1}{AgHg} + \underset{\alpha_1+\alpha_2}{AgCu} + \underset{\varepsilon}{Cu_3Sn} + \ldots + \underset{\eta}{Cu_6Sn_5} + \underset{\gamma}{AgSn}$$

The reaction is not finished yet. After approximately one year, silver–tin phase also vanishes. The tin further diffuses into the silver–copper eutectic slowly and forms more of $\alpha_1 + \alpha_2$ phase. The final reaction is:

$$\underset{\gamma}{AgSn} + \underset{\alpha_1+\alpha_2}{AgCu} + Hg \xrightarrow[\text{one year}]{\text{After}} \underset{\gamma}{AgSn} + \underset{\alpha_1+\alpha_2}{AgCu} + \underset{\gamma_1}{AgHg} + \ldots + \underset{\eta}{Cu_6Sn_5}$$

The difference of reaction in an admixed type and the unicompositional type is that the eutectic silver–copper is absent and the reaction is directly with silver–tin and copper–tin phases.

$$\underset{\gamma}{Ag_3Sn} + \underset{\varepsilon}{Cu_3SN} + Hg \longrightarrow \underset{\gamma_1}{AgHg} + \underset{\eta}{Cu_6Sn_5}$$

Thus, finally the phase formed is Cu_6Sn_5 (η) and no γ_2 phase, i.e. SnHg is formed. The time taken for the reaction is faster in case of unicompositional as compared to admixed type.

The comparative analysis of low copper and high copper silver alloys is given in Table 14.1.

Advantages of silver amalgam

1. *Ease of handling:* The manipulation of silver amalgam, viz. mixing, squeezing and condensing, etc. is very simple and easily understood.
2. *Wide range of applications:* If esthetics is not a major concern, silver amalgam can be given in any class and any size of the cavity. It can also be used as retrograde fillings.
3. *Optimal dimensional changes:* The silver amalgam falls within the range of ADA specification no. 1, which permits $\pm$ 0.2% expansion/contraction during setting.
4. *Physical characteristics:* Physical characteristics of silver amalgam are comparable to enamel and dentin. The compressive strength, hardness, etc. are within optimal limits.
5. *Biocompatibility:* Silver amalgam is biologically stable and accepted. Allergic reactions to components of silver alloy have rarely been reported.
6. Economical.

Disadvantages of silver amalgam

1. Poor esthetics: Even the properly polished silver amalgam restorations are not esthetically acceptable.
2. Marginal degradation or ditching is prominent in low copper or conventional silver amalgams. However, with high copper alloys, marginal degradation or ditching has been minimized.
3. Excessive cavity cutting is required because of its poor edge strength.
4. Tensile strength is less.
5. Silver is a good thermal conductor, so base is required under these restorations.
6. Galvanic current can be produced in association with gold fillings or even in same restoration with non-uniform condensation.
7. Oral lichen planus has also been reported with silver amalgam.

CAVITY PREPARATION

Cavity preparation is defined as "*the mechanical removal of caries and shaping the remaining tooth tissues in such a way so that the tooth/restoration can withstand masticatory forces and be able to prevent subsequent caries after restoration*".

The cavity designs for amalgam restorations are as follows.

Class I

Class I preparation includes cavities in occlusal pits and fissures and also the buccal/lingual pits of posterior teeth and lingual pits of anterior teeth.

A. Cavities in the Occlusal Pits and Fissures

After planning the outline form, the surface is cleaned with water and air. A small round bur is used to enter the deepest or most carious pit, moving the bur parallel to the long axis of the tooth crown. When the bur touches the dentinoenamel junction or slightly into the dentin, the straight fissure bur/diamond point is utilized for further preparation. Conventionally, No. 245 (SS White) carbide bur is used to obtain convergent cavity form along with rounded line

Table 14.1: Comparative analysis of low copper and high copper silver alloys

Low copper	*High copper*
• More mercury is required for amalgamation (53.37%). Solubility of tin in mercury is 170 times more than in copper and 17 times more than in silver.	• Less mercury is required.
• Dominant phase is AgHg, i.e. γ_1.	• Dominant phase is Cu_6Sn_5, i.e. η.
• Corrosion due to γ_2 phase is due to formation of tin phase. oxychloride from tin. Dissolution of this oxide or chloride leads to porosity.	• Cu_6Sn_5 (η) phase is the least corrosion resistant • Corrosion occurs in the form of $CuCl_2$ 3Cu $(OH)_2$.
• Surface tarnish is associated with γ and γ_1.	• Surface tarnish is due to copper-rich phases.
• Low copper alloys can be amalgamated in slow speed and low energy amalgamation.	• High copper alloys require high speed and high energy amalgamation because copper has low solubility in mercury as compared to silver and tin.
• Setting reaction is slow. Early burnishing and finishing is recommended.	• Setting reaction is fast. It can be burnished at first appointment.
• Low copper amalgam has higher value of creep (1–8%).	• Creep is much less (0.1–1.0%.)
• Compressive strength between one hour and 7 days is 150–350 MPa.	• Compressive strength varies between 250 and 500 MPa between one hour and 7 days, especially with unicomposition alloys.
• Tensile strength in 24 hours is 60 MPa.	• Tensile strength is 64 MPa for single composition alloys.
• Dimensional changes are more in low copper alloy; varies from 10–20 µm/cm.	• Much less with high copper alloy; varies from 1–9 µm/cm.

angles. When the pits are equally carious, cavity preparation can be started on any pit. The fissure bur is moved along the fissures maintaining the uniform depth, which is initially 1.5 mm (Fig. 14.3A–D). The preparation at the initial level is kept slightly short of the desired outline form. Again start with the same

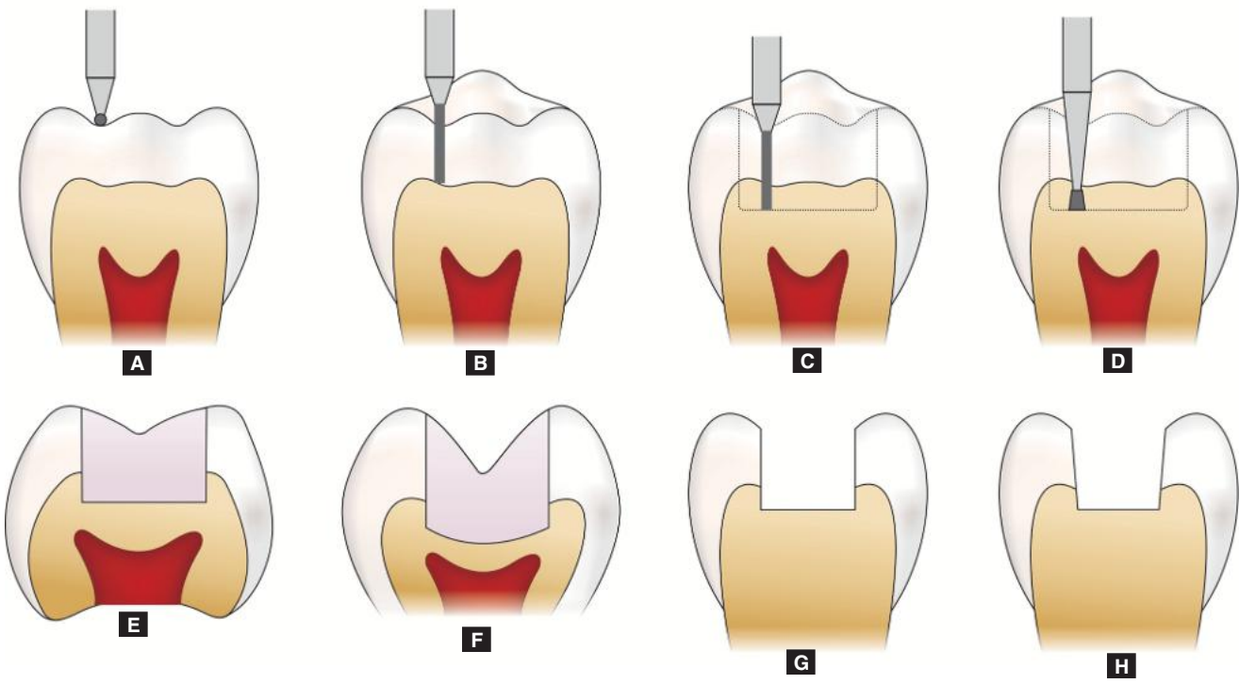


Fig. 14.3: Preparing Class I cavity in occlusal surfaces: **(A)** Deepest or most carious pit is entered with a small round bur; **(B)** When the bur touches the DEJ, No 1 fissure bur is used for further preparation; **(C)** Penetrate fissure bur 0.5 mm below the DEJ; **(D)** Flat pulpal floor is created with inverted cone bur; **(E)** Ideal depth for silver amalgam; **(F)** Modification of pulpal floor when cusps are steep; **(G)** Flat pulpal floor with parallel buccal and lingual walls; **(H)** Flat pulpal floor with 10° divergence of mesial and distal walls

bur, with a light push-pull motion in the fissures to reduce the facial and lingual walls. This way a flat pulpal floor is created with facial and lingual walls almost parallel. However, in teeth with steep cuspal inclines, the pulpal floor can be kept according to cuspal heights, maintaining uniform depth of the pulpal floor (Fig. 14.3E, F).

Extension into the marginal ridges dictates slight tilting of the bur (approximately 10°) mesially and distally to create occlusal divergence at mesial and distal marginal ridges respectively (Fig. 14.3G, H). This procedure prevents the undermining of marginal ridges.

The distance of margins of such an extension to the proximal surface must not be less than 1.5 mm in premolars and 2.0 mm in molars. In case the distance is slightly less, the divergence of the walls can be avoided. However, if the said distance is approximately 1.0 mm, the inclusion of proximal surfaces can be considered (Fig. 14.4A to C).

The buccolingual width of the cavity should be approximately 1.0–1.5 mm or 1/4th of the intercuspal distance. In case, the caries so dictates, the width can be more, but in no case, more than half the cuspal inclines should be involved. In case, where more than half or two-thirds cuspal inclines are involved, the covering of cusps become mandatory.

Alternatively, in spite of covering cusps in such circumstances, the undermined carious lesions are excavated from below the cusps and this undermined area is filled with glass-ionomer cement or composite resin. The rest of the cavity is restored in routine.

All the cavity walls should be joined by definite line and point angles, which are kept rounded (Fig. 14.5A, B).

The cavosurface margins are kept at right angle to the cavity walls. This type of butt joint minimizes marginal breakdown of amalgam (silver amalgam, being brittle, gives way under occlusal stresses at

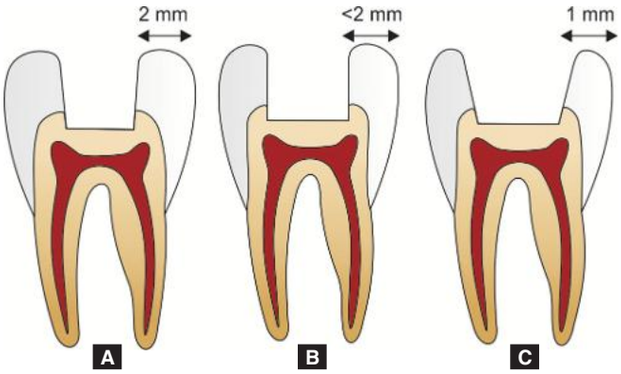


Fig. 14.4: Extension of cavity outline along marginal ridge: **(A)** Adequate marginal ridge and divergence; **(B)** Avoid divergence; **(C)** Inclusion of proximal surface indicated

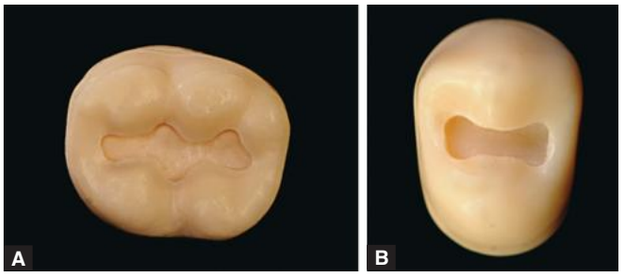


Fig. 14.5: Class I cavity preparation: **(A)** Mandibular first molar; **(B)** Maxillary first premolar

the thinner margins). Bevel/flaring is not created, because the feather edge so produced is liable to fracture. However, in teeth with steep cuspal inclines, the butt joint is compromised and the cavosurface margins are kept at an angle less than 90°. The acute angle in such preparation protects the cusps by minimizing the undermining of enamel. The depth of the cavity is placed 0.5 mm below the dentinoenamel junction. This much depth provides bulk of silver amalgam as well as retention because of elasticity of dentin. When the silver is condensed, the dentin expands and later contract to make a grip of the restoration. *The question arises, if the depth is beneficial both ways (bulk and elasticity of dentin), then why not to extend the cavity 2.0 to 3.0 mm below the dentinoenamel junction. The simple answer is that by doing so the more of dentinal tubules will be in contact with silver amalgam and the alloy being good conductor, leads to sensitivity to hot and cold. It is established that 2.0 to 2.5 mm of remaining dentin is an automatic insulator and the rest can be utilized as an elastic for the benefit of silver amalgam restorations.*

The crossing ridges should be preserved as far as possible (transverse ridge and oblique ridge). If the caries involves more than half the planes of these ridges, the total ridge can be involved. Remember involving these ridges is always detrimental for tooth (Fig. 14.6A, B).

In case, where the caries or the anatomic contours demand the divergent walls other than mesial and distal marginal ridges, the cavity walls up to dentinoenamel junction is kept parallel and the rest can be divergent. The parallelism in the dentinal walls will provide the requisite resistance and retention form to the restoration.

Many a time, caries extends at one or more points below the otherwise flat pulpal floor. Such carious lesions are either excavated with fine excavators or removed with small round burs. These depressions in the pulpal floor are filled with sedative cement, like zinc oxide eugenol or calcium hydroxide. All other features remain the same (Fig. 14.6C, D).

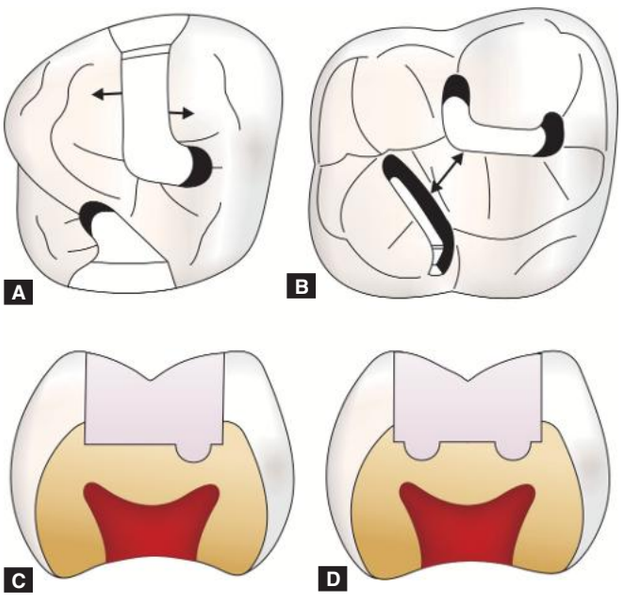


Fig. 14.6: **(A and B)** Preserving transverse oblique ridge in maxillary molars; **(C and D)** Localized extension of the cavity at the pulpal floor

B. *Cavities in the Pits and Fissures on the Occlusal 2/3rd of Buccal and Lingual Surfaces of Posterior Teeth and Lingual Pits of Anterior Teeth*

The pits and fissures on the occlusal 2/3rd of the buccal and lingual walls of the posterior teeth can be involved either individually or in association with the pits and fissures on the occlusal surfaces. In case, where the pits and fissures of only one surface are involved, the decayed portion removed and widened accordingly. The shape is generally oblong or triangular depending upon the extent of caries and the mesiodistal width of the tooth (Fig. 14.7A, B). The walls of the cavity are joined in a continuous fashion. The axial wall is kept flat and the surrounding walls are made slightly convergent towards the cavosurface margins. Preferably, the angle between the axial and the surrounding walls is kept acute. The cervical wall is kept following the contour of cementoenamel junction.

The carious lesions at the buccal and lingual sides are generally connected to the occlusal lesions. This connection may be external or internal. Internal connecting lesions usually undermine the overlying enamel; therefore, have to be involved in cavity preparation.

In such cases, the occlusal portion is prepared as described in Class I. Care is taken to protect the cuspal planes as far as possible, especially, the ones adjacent to the carious grooves, e.g. in case of distopalatal cusp of maxillary Ist molars, since the cusp is small, the cutting is carried out more at the expense of the oblique ridge. In case of distal cusp of mandibular first molar, the cutting is done more at the expense of distobuccal cusp. These precautions will avoid the weakening of cusps.

After preparing the occlusal cavity, keep the straight fissure bur perpendicular to the pulpal floor and move towards the buccal/lingual direction as the caries dictates. Then with the side of tapering fissure bur, the step is prepared, keeping the bur parallel to the buccal/lingual surface of the corresponding grooves (Fig. 14.8).

The axial wall, so produced, is placed in dentin, 1.0 mm inside the dentinoenamel junction. The gingival wall is made 1.5 mm wide and is extended till termination of the buccal/lingual grooves. The axial and gingival walls of the buccal/lingual box are flattened keeping mesial and distal walls parallel. These two walls, mesial and distal, can be kept convergent occlusally in case caries dictates wider gingival walls (Fig. 14.9).

In case, where the apical ends of the buccal and lingual grooves are at the same level as the pulpal floor, the buccal/lingual extension will have no steps. Many a times, the pulpal floor can be deeper as compared to the apical end of these grooves, a

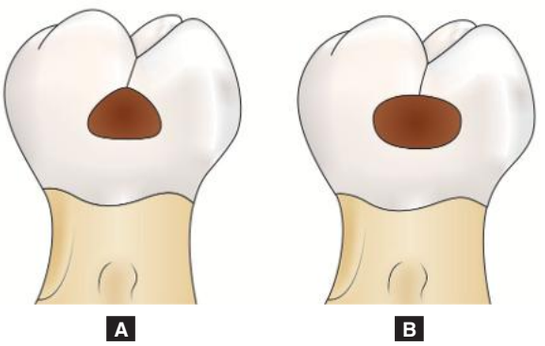


Fig. 14.7: Cavities in the buccal pits of posterior teeth: **(A)** Triangular form; **(B)** Oblong form



Fig. 14.8: Preparation of lingual wall of Class I buccal complex in mandibular first molar



Fig. 14.9: Preparation of mesial and distal walls of buccal complex

condition that is called the reverse step in the occlusal direction. In these cases, the axial wall is created using glass-ionomer, zinc phosphate cement, fast setting zinc oxide eugenol, etc.

The gingival wall meets the external tooth surface at 90° angle (cavosurface margin). The axial wall makes an obtuse angle with the pulpal floor (axio-pulpal line angle) (Fig. 14.10).

The flat gingival wall along with parallel mesial/distal walls is desirable for proper resistance form. However, for retention form, either occlusal convergence of mesial and distal walls is maintained or retention locks are given. These locks are placed in dentin in both mesioaxial and distoaxial line angles. These locks terminate at the level of pulpal floor.

The cavosurface margins of the box as well as the occlusal cavity are finished with enamel hatchets.

Similarly in case of lingual pits in the anterior teeth, the shape of the cavity can be round or oblong. The depth and walls are kept as in buccal/lingual pit cavity preparation (Fig. 14.11).



Fig. 14.10: Class I cavity with buccal complex in mandibular first molar



Fig. 14.11: Cavity on the lingual pit of maxillary lateral incisor

Class II

The outline form of the occlusal part is same as described for Class I; however, outline form of the proximal box is dictated by various factors, viz.:

i. *Extent of caries:* Mostly, the caries are around contact areas; the buccal and lingual walls of the proximal box are kept in self-cleansing areas.
ii. *Convexity of the proximal surfaces:* In cases of convex proximal surfaces, the contact area is comparatively smaller and the extension of the cavity preparation will be minimal towards the embrasures.
iii. *Caries and plaque indices:* The more the caries and plaque indices, the more is the need to extend the proximal walls into self-cleansing areas.
iv. *Masticatory loads:* In case, where the masticatory loads are great, the buccolingual width of the total preparation is kept minimum possible.

The occlusal cavity is prepared keeping the pulpal floor flat, 0.5 mm into dentinoenamel junction and maintaining the parallelism of opposing walls. Position the flat fissure bur perpendicular to the pulpal floor and extend towards the marginal ridge of the proximal area to be involved. The flatness of the pulpal floor is maintained at that level too.

The isthmus width should be as narrow as possible and should not be more than 1/4th of intercuspal distance or 1.0–1.5 mm wide.

Hold the bur perpendicular to the pulpal floor and using light motion, go deep slightly below the contact point and create a trough. This way, initial axial and gingival walls are created (Fig. 14.12A, B). The bur is moved buccally and lingually including the contact area. Care should be taken, if convenient, to leave thin shell of enamel over the preparation to protect the adjacent tooth coming in contact with the bur. Later, this shell can be removed by using enamel hatchets. Alternatively, the adjacent tooth can also be protected by using matrix bands.

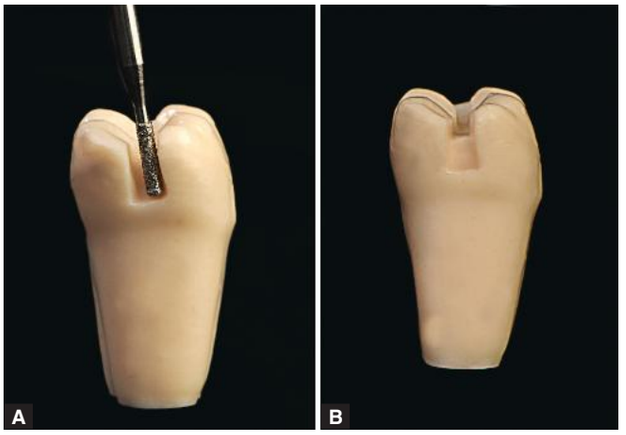


Fig. 14.12: Class II cavity in mandibular first molar: **(A)** Preparation of buccal and lingual proximal walls; **(B)** Prepared cavity (proximal view)

The buccal and lingual proximal walls are kept in embrasures and meet the tangent of the respective cavosurface margin at 90° angle (Fig. 14.13A, B).

In case of maxillary teeth, mostly the molars, the contact area is more buccally placed, i.e. the lingual embrasures are larger than the buccal embrasures. In such cases, extending the buccoproximal wall into the embrasure may lead to excessive cutting of the buccal cusps. To avoid this, a reverse curve is made in the buccoproximal wall, so as to have sufficient amount of dentin in that area and also to achieve butt joint with the cavosurface margins. This curve is towards the axial wall side and the rest is kept straight. The total wall is not curved, since by doing so the silver amalgam restoration will not be at right angle to the cavosurface margins, thereby weakening the restoration.

Such a curve, though mostly given in maxillary molars, can be given in any tooth where the contact area is deviated or more pronounced on one side.

The marginal trimmers are used to bevel or round off the axiopulpal line angle. The gingival wall is beveled 20° towards the cavosurface margin using gingival marginal trimmers. Such an angulation/bevel is given in enamel and if the gingival wall is in

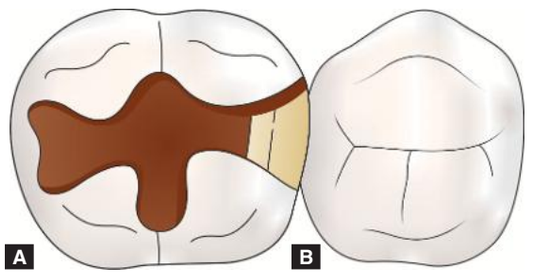


Fig. 14.13: Walls of the proximal box of Class II cavity in the embrasure area

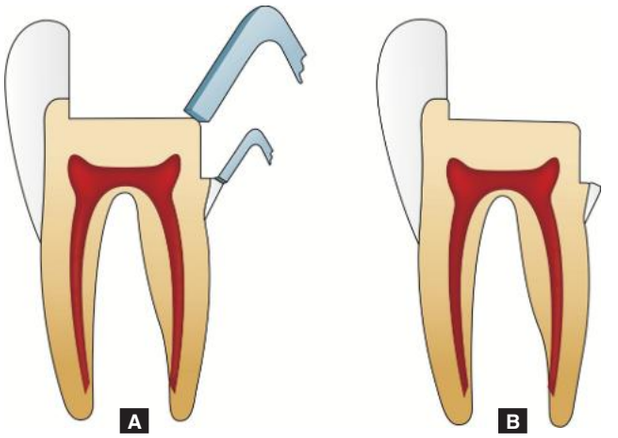


Fig. 14.14: **(A)** Finishing of axiopulpal line angle and gingival cavosurface margin; **(B)** Finished axiopulpal line angle and gingival cavosurface margin

cementum, such an angulation/bevel is not required. The gingival cavosurface margins are trimmed off of the unsupported enamel. The buccogingival and linguogingival line angles are also kept rounded (Fig. 14.14A, B).

Care is exercised to create the gingival wall involving both enamel and dentin. The approximate depth of the gingival wall is 1.0 mm or 1.5 mm, out of which 70% should be in dentin and the rest in enamel. The proximal box walls should have sufficient dentin; facilitates retentive grooves/locks, if need be.

The extension of gingival wall, if in the cementum, the depth should not be more than 1.0 mm (approximately 0.7–0.8 mm).

The proximal box is wider at the gingival area as compared to the occlusal, i.e. the buccal and lingual walls are converging at the occlusal surfaces. (buccolingual width of gingival wall is more than the buccolingual width of occlusal area). This type of occlusal convergence contributes to the retention form and the desirable extension of buccal and lingual proximal walls into the embrasure areas facilitates the restoration to be self-cleansing. The marginal ridge is also conserved by keeping the proximal walls convergent occlusally (Fig. 14.15A, B).

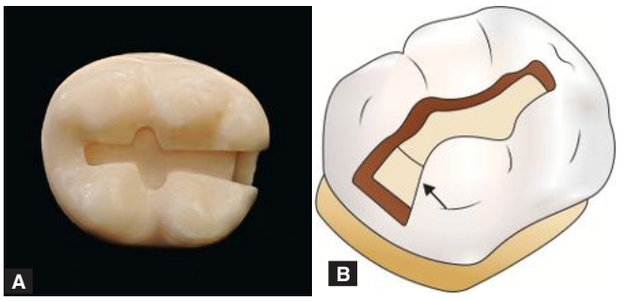


Fig. 14.15: Convergent proximal box walls conserve marginal ridge (arrow)

Primary resistance and retention form is achieved by the following features:

- The gingival and pulpal floor is flat and perpendicular to the long axis of the tooth.
- All walls create 90° cavosurface angle.
- Extension of walls is in such a way that the cusps and ridges have sufficient dentin support.
- Internal line angles are rounded off, especially the axiopulpal line angle, so as to have maximum bulk of silver amalgam at the axiopulpal area and to avoid stress concentration in the tooth.
- Pulpal floor is prepared adequately into the dentin, so as to provide sufficient bulk of restorative material.
- Occlusal convergence of the walls provides primary retention form.

Removal of residual caries at the pulpal floor is accomplished in the same way as described for Class I cavities.

Similarly, caries in the axial wall does not dictate cutting the entire axial wall towards the pulp. A small round bur is utilized to excavate the caries and the area is filled with calcium hydroxide or zinc oxide eugenol cement.

If caries is left in the gingival wall, the small portion can be extended cervically with small round burs leaving the other portion intact.

In case residual caries are left in one or both the side walls of the proximal box, i.e. buccal and lingual/proximal walls, then the extension of the walls is permissible keeping in mind the following points:

- The extension does not cross the line angle of the tooth.
- The wall is not weakened from inside.
- It does not hinder the support of amalgam on the gingival wall.
- The extended area remains accessible.

In case caries still present, cuspal coverage is preferred.

Secondary Retentive Devices for Silver Amalgam Restorations

The commonly utilized secondary retentive devices for silver amalgam restoration are:

1. Slot (horizontal groove)
2. Lock (partial vertical groove)

1. ***Slot (horizontal groove):*** This is a definite depression in the pulpal and/or gingival wall. Slot is prepared with ¼ or ½ round bur depending upon the width required or available, generally 1.0 mm deep and 0.5 mm wide. The slots are always given in dentin (0.2 to 0.5 mm inside the dentinoenamel junction) (Fig. 14.16A, B). The configuration of the slots depends upon the configuration of the gingival floor.

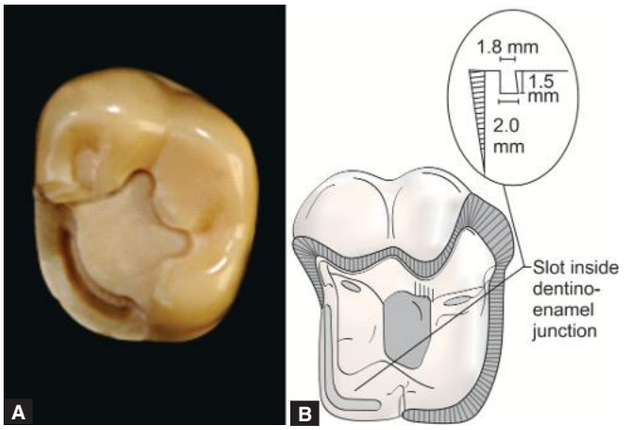


Fig. 14.16: Slot preparation

2. ***Lock (partial vertical groove):*** The locks are generally prepared in proximal box only; however, minor undercuts given in occlusal area serve the same purpose.

The selection of the bur depends upon the required depth of the lock. The bur is moved in axiolingual line angle 0.2–0.5 mm inside the dentinoenamel junction and keeping the pressure both on the lingual and axial side (Fig. 14.17A, B). Remember the pressure should be on both the walls. The pressure only on axial wall can lead to exposure.

Position of axiobuccal and axiolingual line angles is important prior to placement of locks. In shallow line angles, proper amount of dentin is not available and in case of deep line angles, further cutting has the risk of exposure.

It is established that with the use of amalgam with a creep value less than 1.0%, the locks are of no use. (High copper alloys have this creep value.)

Modifications of Class II Cavity

Sometimes the caries involves both the proximal surfaces. The cavity design can be two separate Class II preparation or both united at the occlusal part

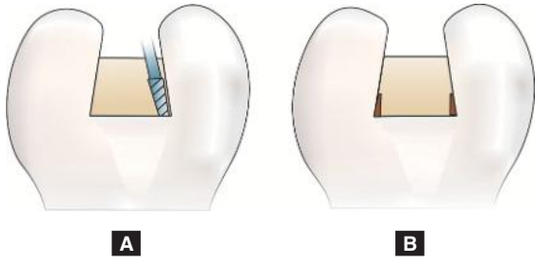


Fig. 14.17: Locks in proximal box: **(A)** Creating lock with appropriate bur; **(B)** Lock created

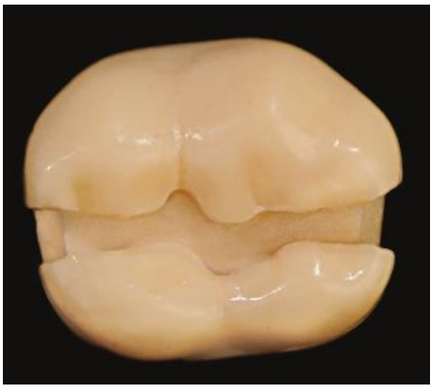


Fig. 14.18: Mesio-occluso-distal (MOD) cavity on mandibular first molar

(MOD) (Fig. 14.18). The union of two proximal boxes at the occlusal level depends upon following factors:

i. Whether the connecting fissures are carious or not.
ii. The remaining thickness of the oblique ridge should not be less than 1.5 mm in maxillary molars.
iii. The oblique ridge or the transverse ridge is not undermined by caries.

In a rare case, the proximal lesions (whether on one surface or both) may have lesions on the buccal and/or lingual surfaces. These lesions can be treated separately, if the carious lesions are separate and the marginal ridge areas are not involved. However, if the caries is undermining the marginal ridges or the remaining width is less, the lesions can be joined as described in Class I cavities (Fig. 14.19A to C).

Occasionally, the proximal lesions are connected to the cervical carious lesions through proximal line angles. In such cases, cervical cavity and proximal lesion is joined, cutting the tooth structure in between (Fig. 14.19D).

Class III

The distal surfaces of canines are the only lesions among anterior teeth, which are filled with silver amalgam.

Resins are usually avoided in the canines because the teeth are subjected to all three types of forces, i.e. compressive, tensile and shear. The anterior component of forces is resisted by the distal surfaces of canines; subsequently, continuous rubbing the distal side of the tooth. The resin restorations, because of their poor wear resistance, may wear off easily.

The outline of cavity is planned. A small round bur (No. ½ or 1) is used to enter the carious lesion from the marginal ridge lingually. The direction of the bur is maintained perpendicular to the lingual surface of the tooth. The bur is moved 0.5 mm inside the dentino-enamel junction. The labial wall should have intact enamel as far as possible (Fig. 14.20A). The lingual

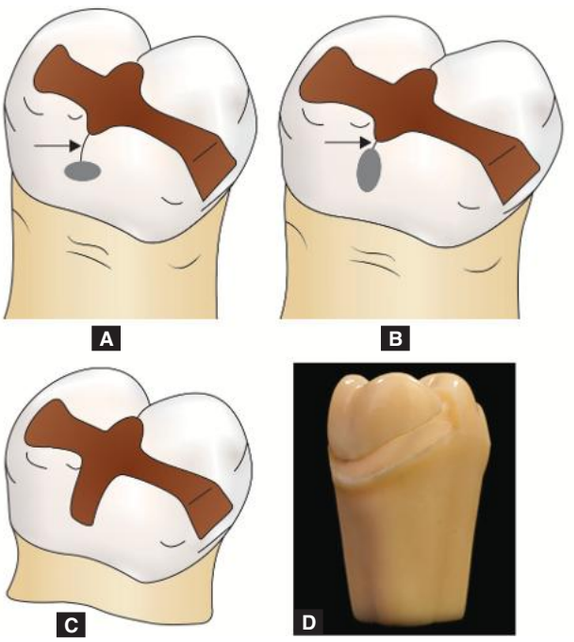


Fig. 14.19: Extension of cavity margins towards buccal/lingual carious lesion (arrow): **(A)** Adequate marginal ridge; no need to join the preparation; **(B)** Thin marginal ridge; join the preparation; **(C)** Joined preparation; **(D)** Class II + Class V joined cavity



Fig. 14.20A: Class III (distal canine): Maintaining intact labial wall enamel

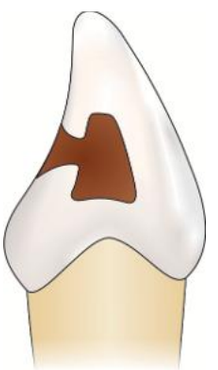


Fig. 14.20B: Class III cavity preparation; no lingual proximal wall

outline in such cavity preparation blends with the incisal and gingival margins; no lingual proximal wall (Fig. 14.20B).

The preparation can extend lingually to create a lingual lock, if need be. The lingual portion is extended incisally as well as gingivally.

The lingual extension in no case is extended beyond the imaginary half of the lingual surface. The isthmus between the lingual and proximal preparations is kept approximately 1.5 mm (Fig. 14.21).

All cavosurface angles are at 90° to the margins. The labial, incisal and gingival walls should meet the axial wall at right angle. The lingual wall, if present, may be making an obtuse angle with the axial wall (Fig. 14.22A, B).

Preferably, the contact point is not disturbed, until and unless dictated by caries. The drifting of canines distally is very common; therefore, proper contact will prevent or minimize the movement of the tooth.

Resistance form is provided by:

- Cavosurface angles of 90°
- Definite internal line angles
- Sufficient bulk of the material, i.e. adequate depth

Retention form is provided by:

- Preparing a gingival and incisal groove or undercut (Fig. 14.23A to C). The gingival groove is placed at the axiogingival line angle using small round/inverted cone bur. The groove is usually 0.2 mm wide and 0.3 mm deep in dentin only.

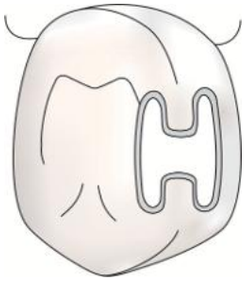


Fig. 14.21: Class III preparation (diagrammatic)

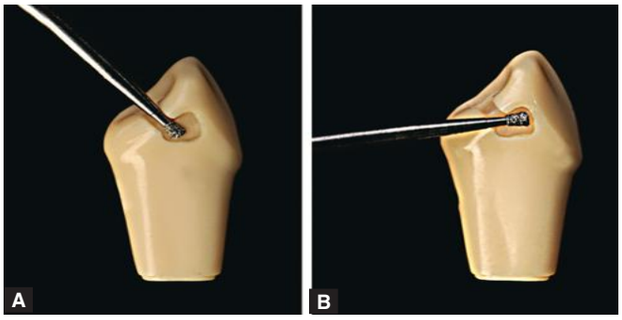


Fig. 14.22: Class III cavity on distal side of maxillary canine: **(A)** Gingival wall preparation; **(B)** Labial wall preparation

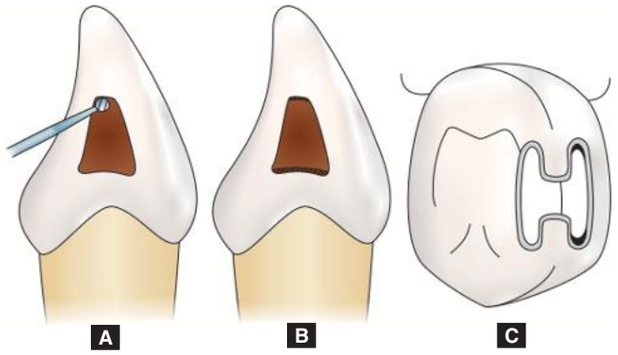


Fig. 14.23: Incisal and gingival undercut/groove



Fig. 14.24: Palatal dove tail preparation for Class III cavity in maxillary canine

- The incisal groove is prepared in the incisoaxial line angle or inciso-axio-labial point angle, as in gingival groove.
- The grooves can also be placed on the palatal dove tail of the preparation for added retention (Fig. 14.24). The cavity margins are smoothened to have 90° cavosurface margin.

Class IV

Silver amalgam is not indicated in Class IV cavities because of its poor edge strength and also poor esthetics.

Class V

The cervical caries/abrasion and erosion lesions are very common. These lesions can be entirely in cementum or partially in enamel and cementum. The labial and lingual surfaces of posterior teeth can be restored with silver amalgam while the labial surfaces of anterior teeth can be restored with composite resins.

These restorations are subjected to functional stresses during protrusive and lateral movement of the mandible. These types of forces tend to dislodge

the restoration buccally and create a minor gap at the occlusal/cervical surface of the restoration.

This gap is very mild and usually creates no problem. However, with repeated movements, the restoration may get dislodged.

A flat fissure bur of appropriate size is used to enter the center of the lesion at the depth of 0.5 mm inside the dentinoenamel junction. By keeping the bur perpendicular to the long axis of the tooth, the bur is directed mesially and distally and then occlusally and gingivally (Fig. 14.25A, B). The mesial and distal extension of the preparation should be up to the line angles of the tooth until and unless the extension beyond the line angles is dictated by caries. The gingival wall is contoured according to the cementoenamel junction and the occlusal wall is, by and large, straight. The axial wall is also contoured mesiodistally following the outer contour of the tooth. The depth of the gingival wall is usually 0.75 mm and the occlusal wall is 1.25 mm.

All the four walls of cavity preparation, i.e. mesial, distal, occlusal and gingival, diverge outward. The gingival wall, if placed in cementum, is kept straight. A small round/inverted cone bur (No. ¼ or ½) is used to prepare the retention grooves, one in axioocclusal line angle and the other in axiogingival line angle (Fig. 14.26A to C).

In rare cases, if the cavity is extensive, the groove can be given all around the four walls of the cavity (Fig. 14.27A, B).

The cavosurface margins are kept at 90° for the proper resistance form.

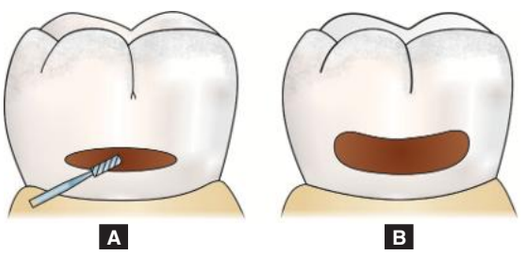


Fig. 14.25: Preparing Class V cavity: **(A)** Entry into the lesion with fissure bur; **(B)** Prepared cavity

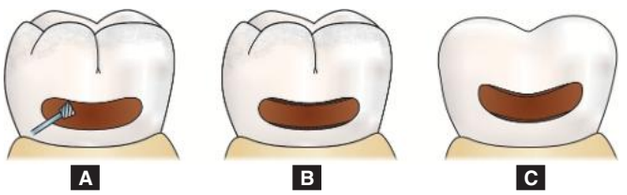


Fig. 14.26: **(A)** Bur directed occlusally to prepare groove; **(B)** Preparing grooves on occlusal and cervical walls angles; **(C)** Preparation completed

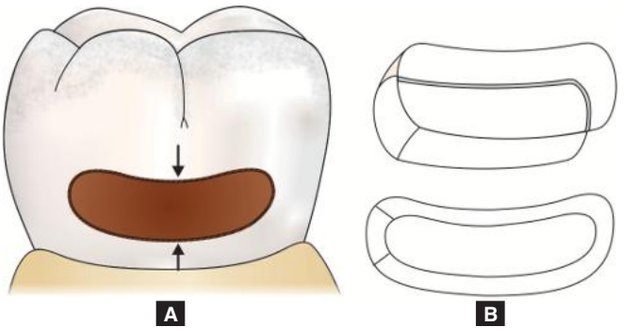


Fig. 14.27: Groove all around the four walls of Class V cavity

Modifications of Class V

In rare instances, the lesion/cavity can be on the line angles of posterior teeth.

The cavity preparation is as for class V cavities, creating a round or oblong shape, as dictated by caries.

The depth is kept 0.5 mm into the dentinoenamel junction. The cavosurface line angles are kept at 90° to achieve butt joint of the silver amalgam with the cavity walls.

A small retentive groove is given on the occlusal and gingival wall of the preparation with ¼ or ½ round bur to achieve retention form, wherever necessary.

Class VI

The lesions on the incisal edges are not restored with silver amalgam; however, the cusp tips (carious or hypoplastic) can be restored with silver amalgam.

The center of lesion is penetrated with small tapering fissure bur and extended all around to involve the caries (Fig. 14.28A to C). A 90° cavosurface angle is usually given and if the cuspal planes are steeper, the cavosurface angle can be modified accordingly. A small retentive groove using ¼ round/inverted cone bur can be created all around the internal line angle. The shape of the cavity is usually round.

MANIPULATION OF SILVER AMALGAM

Adequate cavity preparation need proper manipulation of silver-mercury for successful restoration.

The sequence of manipulation is as follows.

a. Selection of Alloy

The features considered for selection of alloy are:

i. *Type of alloy:* Among the various types of alloys available, the choice depends on the properties and their composition like:
   - Size and shape of particles—dictate the amount of mercury required.

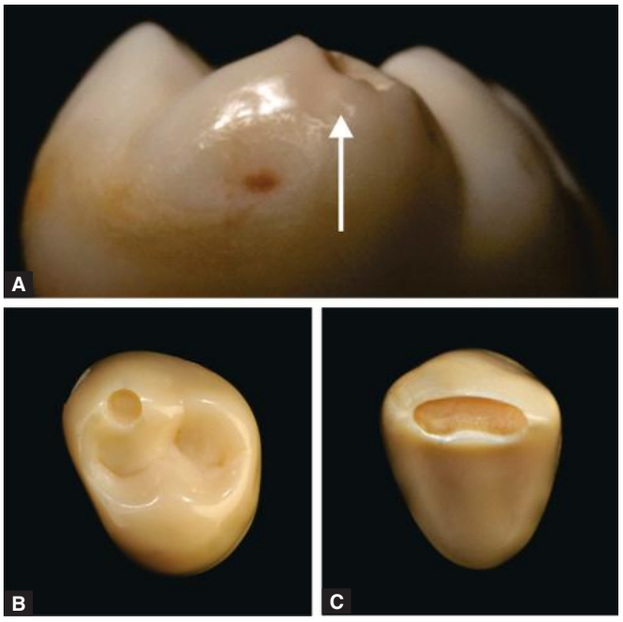


Fig. 14.28: Class VI cavity: **(A)** Caries at cuspal tips; **(B, C)** Prepared cavities

- Alloys containing zinc or without zinc
- Percentage of copper in given alloy, i.e. high copper (>13%) and low copper (<6%)

ii. *Anatomic position of restoration:* For restoration subjected to heavy occlusal forces, an amalgam with high resistance to marginal fracture is desirable.

iii. *Feasibility of isolation:* The feasibility to isolate the tooth shall determine the choice between zinc-containing or zinc-free alloys.

iv. *Availability of working time in clinic:* Certain patients should be disposed off early; quick setting type of alloy is required in such cases.

v. *Extent of the cavity:* Wider cavities have more chances of deterioration; alloys with low creep values are required for such cases.

b. Proportioning of Alloy and Mercury

Mercury must wet the alloy particles; however, wetting is dependent on a number of factors, like alloy composition, surface condition, particle size and shape. The more quantity of mercury is generally required with hand mixing, while low quantity of mercury is required for mechanical mixing. The spherical particles require less mercury (40%) as compared to lathe cut particles (45%). In routine, alloy/mercury ratio is kept as 5:8 or 5:7, although Eames has preferred 1:1 ratio. If extra mercury is taken, there will be more chances of formation of weaker γ_2 phase and if less mercury is taken, it might not sufficiently wet the alloy particles, leading to porous amalgam.

c. Dispensing of Alloy and Mercury

Alloy can be dispensed either by weight or by volume. Mercury is commonly dispensed by volume. Bottle should be held vertically and kept at least half full to ensure uniform spills.

Semiautomatic dispensers are commonly employed, which also carry out trituration. These dispensers have two containers, containing alloy and mercury. On pressing the button, given amount of mercury and alloy is released.

Encapsulated alloy/mercury is also used. In a mechanical mixer, the membrane is broken, as the mixer starts vibrating.

d. Trituration

Trituration can be carried out by hand or by mechanical amalgamators. Hand trituration is performed in glass mortar and pestle, preferably with a roughened surface. Mechanical mixing is carried out in semiautomatic or fully automatic amalgamator.

The factors controlling quality of trituration are:

i. *Time of trituration:* Varies (10–30 seconds); following features are significant:
- Higher for low copper alloys and less for high copper alloys
- Type of amalgamator
- Pestle of different weight and size
- Increasing size of mix increases the mixing time

ii. *Speed:* Mostly depends upon the unit. In case of mechanical amalgamators, the speed is set by the manufacturer. In hand mixing, the speed varies from operator-to-operator.

iii. *Force applied during trituration/amalgamation:* Depends upon weight of pestle (more weight more is force), and the design of the pestle (big size, more force).

Excessive force during trituration may lead to splintering of alloy particles leading to weak matrix. During trituration, following precautions are helpful:
- Mortar should be rested on a firm base
- Uniform pressure should be applied.
- Time should be controlled.
- Surface texture of pestle and mortar should be rough.

The consistency of the mix is categorized as; normal, under mix or over mix.

i. The *normal* mix consistency is shiny, warm, homogenous and has good strength.

ii. The *under mix* consistency is dull in appearance and inconvenient to manipulate during insertion.

iii. The *over mix* consistency is soupy, difficult to remove from the pestle and has low strength.

Advantages of mechanical amalgamator:
- Mix is uniform.
- Trituration time is less.

- No need to squeeze excess of mercury, since the alloy/mercury ratio is proportioned by the manufacturer.
- Reduces the risk of environmental mercury contamination.

e. Mulling

After hand trituration, mulling is carried out by taking mix in hand and rubbing between fingers.

After mechanical trituration, mulling is carried out by rotating the mix in a pestle-free capsule. It gives mix a cohesive form.

After mulling, the mass is taken in a muslin cloth and squeezed properly to remove excess mercury, especially in hand trituration. This squeezing should not be performed in an open environment of clinic, as it can lead to increased mercury vapor content in the atmosphere.

f. Condensation

The mix is carried into the prepared cavity (Fig. 14.29) and compacted to attain a dense mass.

Condensation helps in:

- Development of a continuous matrix.
- Removal of excess of mercury, minimizing porosity.
- Proper adaptation of the material to all parts of the cavity walls.
- Better adaptation of the incremental layers of amalgam.

Operator should condense the mass as early as possible after trituration. If the time lapse between trituration and condensation is more than 3 to 4 minutes, then the mix should be discarded.

The condensation is carried out by following methods, viz.:

i. Hand condensation
ii. Mechanical condensation
iii. Ultrasonic condensation

i. *Hand Condensation*

Various shapes of condenser are available to suit the cavity configuration (Fig. 14.30A, B). Pressure, direction of condensation and the size of increment regulate condensation.

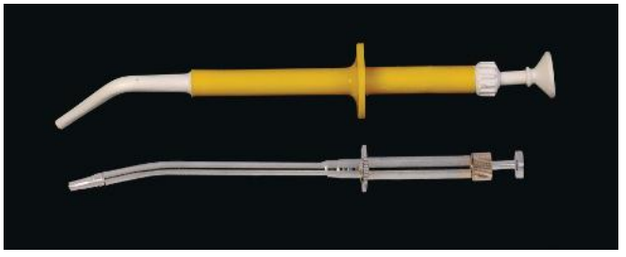


Fig. 14.29: Types of amalgam carrier

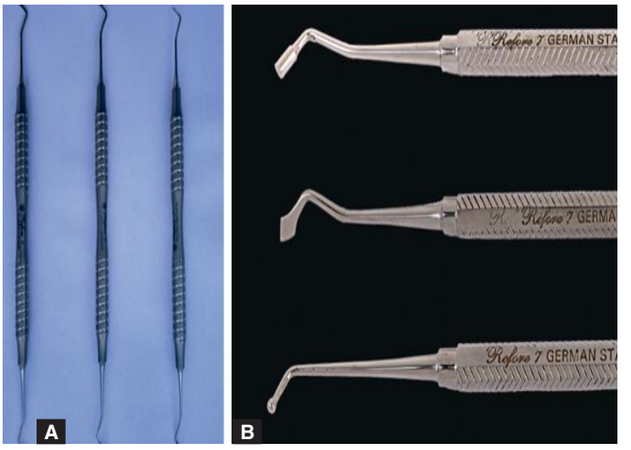


Fig. 14.30: **(A)** Shapes of condensers; **(B)** Amalgam condenser, carver and burnisher

The condensation should preferably start from center to periphery.

The increments should be small at one time. Large bulk of increments facilitates air entrapment and leads to a porous and a weak restoration (Fig. 14.31A, B). The cavity is overfilled slightly, which help in burnishing and carving (Fig. 14.31C).

Condensation pressure in the range of 4 to 8 lb is regarded as most appropriate.

It is established that high condensation pressure substantially reduces the residual mercury content.

Higher condensation pressure leads to close packing of the mass, so the residual mercury rises on the surface, which can be removed during burnishing and carving.

After proper condensation, the surface of restoration becomes shiny. This is due to accumulation of residual mercury at the surface of restoration.

Should condensers be smooth or serrated, remained controversial. Authors favouring serrated condensers are of the view that serrations make the surface of increment rough so that when next increment is added,

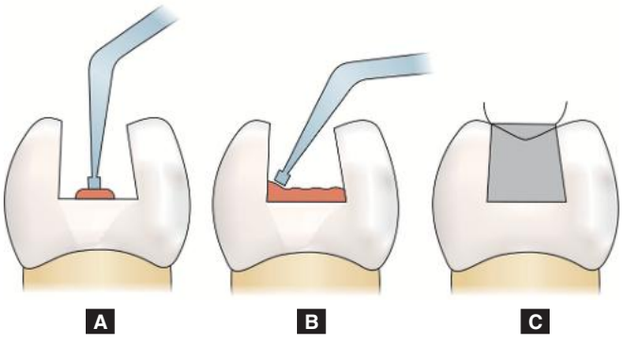


Fig. 14.31: **(A)** Condensation of silver amalgam: **(A)** At the center; **(B)** At the periphery; **(C)** Overfilled cavity

mechanical bonding would take place. Authors favouring smooth condensers are of the view that mechanical retention is of least importance in packing various increments of amalgam mix because bonding occurs due to residual mercury, which occurs at the surface of each increment.

ii. *Mechanical Condensation*

Various shapes and sizes of mechanical condensers are available. The mechanical condensers may apply high loads, damaging the tooth and even lead to cuspal fracture.

iii. *Ultrasonic Condensation*

Ultrasonic condensers are not recommended because these produce local heating of amalgam leading to release of mercury.

g. Finishing and Carving

The objective of finishing and carving is to remove mercury-rich layer on the surface and re-establish the contact with the opposing dentition. The alloy mass should be properly hardened before starting carving. If the mass is not set properly or is still plastic, then initiation of carving leads to pulling out of amalgam from the margins.

When scraping silver amalgam with carver, a ringing sound appears or heard that is taken as a guide for appropriate time of carving.

Carving should be carried out by keeping half of the blade of carver on tooth structure and half on the restoration following incline plane of each cusp.

Burnishing is a process in which a smooth, rigid instrument is used to smoothen the restoration. There are conflicting views as to what should be carried out first, carving or burnishing.

If carving is carried out before burnishing, the effect of carving is lost. If burnishing is carried out before carving, then rough surfaces are produced.

This has led to the concept of pre-carve and post-carve burnishing.

Pre-carve burnishing is carried out before carving, which provide smooth margins and shapes the contours and curvatures (Fig. 14.32).

Post-carve burnishing: After carving, the rough surface so produced is smoothened by final burnishing. At this stage, the mass is hard/set enough to prevent any disturbance of anatomy formed by carving.

FAILURES OF DENTAL AMALGAM

The success of amalgam restoration depends upon many variables, from cavity preparation until the restoration is polished.

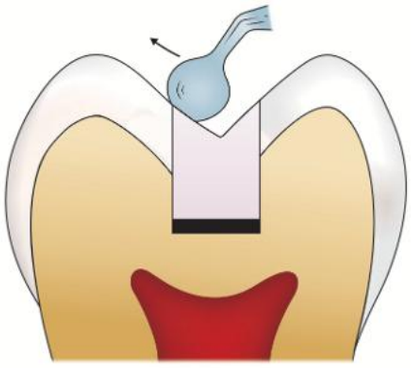


Fig. 14.32: Pre-carve burnishing helps in removing excessive mercury from overpacked amalgam. Always move ball burnisher from central groove to the margin using firm pressure

The types of failure (Fig. 14.33A to D) in an amalgam restoration are:

- At visual level
  - Secondary caries
  - Marginal fracture
  - Bulk fracture
  - Tooth fracture
  - Dimensional change
- At the microstructural level
  - Corrosion and tarnish
  - Stresses associated with masticatory forces
- Pain following amalgam restoration
- Pulp and/or periodontal involvement

Failures in an amalgam restoration are attributed to:

I. Faulty cavity preparation
II. Poor matrix adaptation
III. Faulty amalgam manipulation
IV. Postoperative pain

I. Faulty Cavity Preparation

Improper cavity preparation leading to recurrence of caries and/or fracture is the greatest single factor responsible for failure.

The features of failure during cavity preparation are:

a. *Inadequate occlusal extension:* Inadequate extension to include pits and fissures increases the chances of caries recurrence, especially in patients with high caries index.
b. *Inadequate extension of the proximal box:* If the proximal box walls are not adequately extended into the embrasures; being difficult to clean, may predispose to secondary caries.
c. *Overextension of the cavity preparation walls:* The ideal faciolingual width of the cavity preparation for amalgam should be 1/4th the intercuspal distance. If the cavity preparation extends to 2/3rds of the

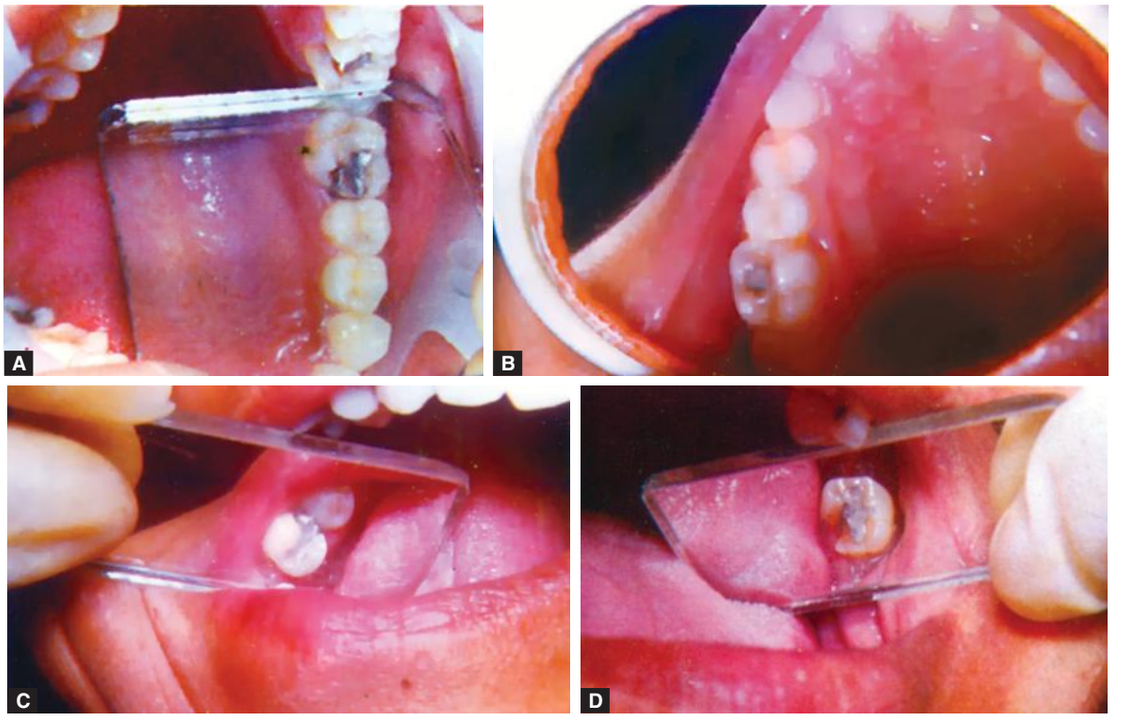


Fig. 14.33: Failures of amalgam restoration: **(A)** Ditching; **(B)** Secondary caries; **(C)** Fractured amalgam; **(D)** Fractured tooth

intercuspal distance, cusps become susceptible to fracture.

d. *Inadequate depth:* Amalgam cavity preparations should have a minimum depth of 1.5 mm to provide the bulk and resistance to fracture.
e. *Curved pulpal floor:* If the pulpal floor of the cavity preparation is not flat but curved, the restoration produces a wedging effect, thus increasing the chances of the fracture of tooth. Flat pulpal floor should also be provided around the excavation site of caries.
f. *Acute/obtuse cavosurface area:* Butt joints should be created, especially in those regions where occlusal stresses are to be encountered. If the cavosurface angle is acute, there are chances of fracture of the tooth margins; whereas, if the cavosurface angle is obtuse, the marginal amalgam is likely to collapse under occlusal stress.
g. *Sharp axiopulpal line angle:* The axiopulpal line angle as well as internal line angles and point angles should be rounded, otherwise may lead to concentration of stresses and fracture of the tooth or restorative material.
h. *Inadequate isthmus width:* Occasionally, fracture may be seen at the isthmus area; narrow isthmus relative to the rest of the cavity preparation lead to fracture.
i. *Undermined marginal ridge enamel:* When the mesiodistal extension of the cavity is extensive, it can cause fracture because of the undermining of the mesial and distal marginal ridge enamel.
j. *Improper retentive devices:* Retentive devices should be prepared entirely in dentin without undermining the enamel.
k. *Use of low speed for cutting:* The operator should use high speed rotary instruments, with intermitting cutting and adequate cooling to minimize the postoperative pain.

II. Poor Matrix Adaptation

The proper contacts and contours in a restoration can be obtained by a properly applied matrix. Instability of a matrix results in a distorted restoration, gross marginal excesses and an uncondensed soft amalgam. The cervical excesses can irritate the periodontium, gradually and progressively destroying the periodontium. An effective condensation cannot be attempted against a poorly stabilized matrix, which will result in a soft amalgam filled with voids.

Premature removal of matrix band may lead to restoration fracture.

III. Faulty Amalgam Manipulation

Successful restoration can be achieved when variables of amalgam manipulations are kept under control.

The variables are as follows:

a. ***Trituration:*** Both undertrituration and overtrituration can lead to failures of amalgam restoration. Undertrituration leads to soft powdery

non-coherent mix; whereas, overtrituration may break the already forming matrix.

b. ***Condensation:*** Condensation can be carried out either manually or mechanically. Small increments should be used rather than large increments to ensure proper condensation. Small plugger may punch holes in the amalgam; whereas, large plugger may not condense the amalgam in all corners. Condensation pressure should be adequate. Mechanical condensation may provide restorations that have slightly higher compressive strength; however, should be used with caution to avoid fracture of enamel margins.

c. ***Contamination:*** Contamination of the amalgam mix with moisture during trituration, mulling and condensation weaken amalgam restoration, especially with zinc-containing alloys. There occurs delayed expansion; subsequently, failure of restoration.

d. ***Finishing and polishing:*** The amalgam should be finished gently. The thin flakes/spur-like overhangs of amalgam can fracture from the restoration, leaving susceptible crevices in vulnerable areas of the tooth surface.

Over carving the restoration to create normal, deep anatomic features should be avoided. An over carved restoration will reduce the thickness of amalgam and increase chances of fracture.

Silver amalgam should be polished judiciously. When temperatures above 65°C are generated, mercury is released from the amalgam, leading to defective restorations.

IV. Postoperative Pain

The postoperative pain can be due to:

- *Hyperocclusion:* Leads to inflammation of the apical periodontium.
- *Cracks in tooth:* Such cracks cause pain during chewing, because of expansion/contraction of tooth structure with every bite.
- *Galvanism:* The adjacent/antagonist dissimilar metal restorations lead to galvanism, and also poorly condensed silver amalgam (variation in silver concentration at different areas of the same restoration, also leads to galvanism).
- *Delayed expansion:* Peculiar to zinc-containing alloys (expands and causes pain).
- *Inadequate pulp protection:* Amalgam is a good conductor of heat. If a base is not given, heat conducted to the pulp may incite pain.

GALLIUM ALLOYS

The toxic effects of mercury coupled with problems of mercury hygiene led the development of mercury-free alloys (gallium alloy).

Composition	
Powder	
Silver (Ag)	60%
Tin (Sn)	25%
Copper (Cu)	13%
Palladium (Pd)	2.0%
Liquid	
Gallium (Ga)	62%
Indium (In)	25%
Tin (Sn)	13%
Bismuth	Traces

The alloys of gallium are mixed and condensed as silver amalgam using almost the same instruments. It sets in reasonable time and possesses strength, diametrical stability and corrosion resistance equal to or even greater than silver amalgam.

- Compressive and tensile strength increases with time comparable with silver amalgam.
- Creep values are as low as 0.09%.
- Sets early, so polishing can be carried out the same day.
- Expand after mixing, therefore, provides better marginal seal.

The physical properties of high copper silver alloys and gallium alloys are summarized in Table 14.2.

Table 14.2: Comparative properties of high copper silver alloys and gallium alloys

Alloy	*Creep %*	*Compressive strength (after 6 hours)*	*Setting contraction/ expansion (%)*
High copper silver alloy	1.04 ±0.06	370 MPa	– 0.05
Gallium alloy	0.09 ±0.03	350 MPa	+ 0.39

Limitations

The problem of setting expansion, though considered good initially, later proved to be deleterious. The gallium alloys could not be used in larger restorations as the expansion led to fracture of the weakened cusps.

The creep values are best suited for gallium alloys. Even high copper alloys which are γ_2-free still exhibit significant creep value. Gallium alloys exhibit negligible creep value.

The compressive strength is adequate; therefore, it can be given in stress-bearing areas. However, its manipulation is difficult. Since these alloys are sticky, their condensation into the cavity is time consuming. This also creates problem with removal of matrix

bands. The cleaning of instrument tips and carriers is also difficult and time-consuming.

MODIFIED SILVER RESTORATIONS

a. Bonded Amalgam Restorations

To improve marginal adaptability of silver amalgam, various bonding agents have been tried.

Superbond (META/methyl methacrylate) and Panavia (BisGMA phosphonated ester) were used earlier.

Later, amalgam bond with HPA (Parkell), all bond 2 (Bisco), Optibond 2 (Kerr), Panavia 21 (Kuraray) (Fig. 14.34), Clearfil Linerbond 2 (Kuraray), Scotchbond MP (3M), etc. were successfully used.

Indications

- Situations that warrant auxiliary retention, reinforcement of remaining tooth structure, conservative preparations and improvement of marginal seal.
- Extensively carious posterior teeth.



Fig. 14.34: Panavia 21

- Teeth with low gingivo-occlusal height.
- Used as a core under a cast crown.

Advantages

- Permits conservative cavity preparations.
- Reinforces tooth structure.
- Eliminates the use of retention pins.
- Decreases the incidence of marginal fracture.
- Provides a bond at the tooth–restoration interface; minimizes microleakage, recurrent caries and post-operative sensitivity.
- Allows biologic sealing of the pulpodentinal complex.
- Cost-effective.

Disadvantages

- Technique sensitive.
- No sustained effects of amalgam bonding when subjected to thermocycling.
- Hydrolytic stability of the bond is questionable over a prolonged time period.

Manipulation

The cavity form in bonded amalgam restorations is conservative; however, adequate form of resistance should be provided.

The bonding agent to be chosen under amalgam should preferably be chemically cured. Amalgam bond, which utilizes the 4-META system, uses a solution of 10% citric acid and 3.0% ferric chloride to remove the smear layer and demineralize the dentin surface. A primer is subsequently applied on the conditioned dentin followed by a self-curing 4-META system.

The procedure for a bonded amalgam restoration is illustrated using All Bond 2 (Bisco) resin (Fig. 14.35 A to C).

The tooth restoration interface in a bonded amalgam restoration is characteristically composed of the tooth, intervening adhesive resin and the amalgam. The tooth-resin bond is similar to the tooth–resin bond in composite restoration.



Fig. 14.35: Bonded amalgam restoration: **(A)** Carious lesion on mandibular left first molar; **(B)** Bonding resin applied on cavity preparation; **(C)** Completed restoration

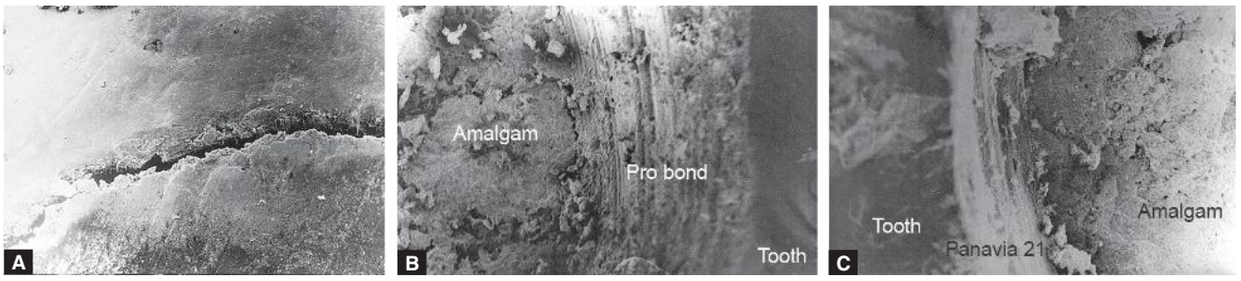


Fig. 14.36: Scanning electron microscopic picture of: **(A)** Fresh amalgam–tooth interface; **(B and C)** Amalgam–resin–tooth interface



Fig. 14.36D: Scanning electron microscopic picture of old amalgam–tooth interface

The scanning electron microscopic picture of fresh amalgam–tooth interface, amalgam–resin–tooth interface is shown in Fig. 14.36A to C and old amalgam–tooth interface is shown in Fig. 14.36D.

The attachment between the adhesive resin and the tooth structure is generally stronger than that between the adhesive resin and amalgam; hence, the weak link in amalgam–resin interface, i.e. an adhesive failure rather than cohesive failure in the resin is more likely at the interface.

The success and failure of these restorations cannot be based solely on the adhesives, but other factors like cavity preparation, type of alloy used, amalgam condensation, occlusion and the patient's dietary and hygienic habits, etc. also play a significant role.

Ability to maintain adhesive bond over a long period of time under hydrolytic environment still remains questionable. Laboratory experiments have failed to show any added advantages of bonded amalgam restorations when the latter were subjected to cyclic loading and thermocycling tests.

b. Fluoridated Amalgam

Fluoride, being cariostatic, has been tried in amalgam to deal with the problem of recurrent caries; however, fluoride is not delivered long enough to provide maximum benefit. An anticariogenic action of fluoride amalgam is explained by its ability to deposit fluoride in the hard tissues around the fillings, increasing the fluoride content of plaque and saliva, subsequently affecting remineralization. However, this effect lasts for less than a weak.

AMALGAM RESTORATIONS AND ORAL ENVIRONMENT

Amalgam restorations are subjected to chemical, biological, mechanical and thermal forces in the oral environment. The features, such as biochemistry of adjacent tissues, biofilm on the amalgam surfaces, galvanism, abrasion, etc. affect silver restoration.

Usually, the environment for external surface of amalgam fillings is, saliva. At the interface between filling and tooth structure, the environment is provided mainly by leakage of fluids from the oral cavity. The fillings are covered initially with acquired pellicle, formed by salivary glycoproteins. This film affects the transport of ions. These modified metal organic complexes along with production of metabolites of micro-organism are potentially significant factors in the corrosion processes (Fig. 14.37).

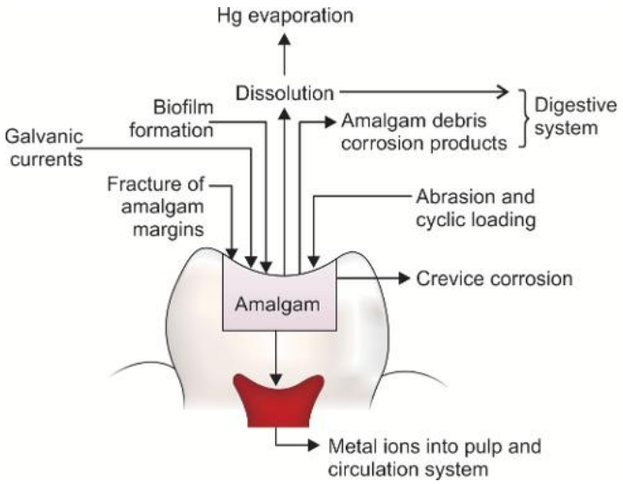


Fig. 14.37: Interactions between amalgam restoration and the oral environment

The effects of oral environment can be as follows.

a. Biodegradation

The electrochemical reactions constitute one of the major interactions between dental amalgam and the oral fluids. The cause of degradation is the mechanical forces to which the fillings are subjected. There can be formation of biofilm, release of mercury from the amalgam or even wear and fracture of the restoration.

b. Corrosion

Corrosion implies electrochemical reactions that result in degradation of the structure and mechanical properties.

The most severe corrosion takes place in localized corrosion cells, such as pit and crevices. Because of the localized chemical changes within these occluded cells, the protective film is not stable, leading to corrosion. If the amalgam restoration comes into contact with another metallic restoration, its electrode potential and the corrosion rate changes. The corrosion is accelerated.

A restoration is a very complex electrode with different parts of the surface corroding under different conditions. The rate at which corrosion occurs varies with different locations. Corrosion decreases the strength of the dental amalgam; may be up to 5.0% or more.

c. Tarnish

The discoloration of the amalgam restorations with the formation of silver and copper sulfides is known as tarnishing. Thicker tarnish layer may result from accumulation of complex deposits of corrosion products and substances from the environment. Tarnish does not lead to loss of strength of the tooth.

d. Wear and Fracture

With the functional forces, the surface of the filling and the enamel is abraded, referred to as wear. In case the load is abnormal, it may lead to ditching or even fracture of the restoration.

e. Release of Amalgam Components

The process of corrosion results in metallic and nonmetallic corrosion products. Most of the soluble metal ions are released into the oral fluids, while a substantial part of solid corrosion products remain within the amalgam structure. Some of these products are continuously removed by abrasion. Amalgam debris occasionally get embedded into the gingiva in the form of amalgam tattoo, releasing the element into the tissue fluids.

f. Release of Mercury

The release of mercury from dental amalgam restoration by dissolution and evaporation do not seem to cause substantial degradation of either appearance or function of restorations. Usually the mercury release is prevented or minimized by a protective film of oxide on the surface. The mechanical removal of the film by abrasion causes substantial increase in the evaporation rate. In the oral cavity, during partial removal of the film by chewing or brushing, the atomic mercury does not evaporate but dissolve in fluids. The dissolved mercury then either evaporates from the liquid solution or is oxidized to an ionic form (Fig. 14.38).

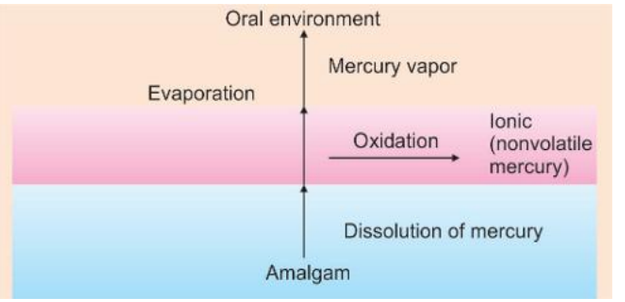


Fig. 14.38: Dissolution of mercury in oral environment

MERCURY AND ITS MANAGEMENT

Mercury has been used as a component of dental amalgam since long. Mercury vapors in atmosphere are considered to be toxic, therefore, management of free mercury in dental operatory is of paramount importance.

Mercury exists in three chemical forms; viz. (i) elemental mercury, (ii) inorganic mercury (salts of mercury), and (iii) Organic mercury (organomercurials).

Elemental mercury is most volatile out of three forms. Inorganic mercury exists as a mercurous or divalent mercuric salts. Salts of mercury can be acutely toxic and irritating. Organic mercury is not a hazard in dentistry, but can be an environmental hazard.

Mercury Exposure in Oral Cavity

Mercury from dental amalgam is released in two forms, i.e. mercury vapors and mercuric ions.

Mercuric ions (Hg^{++}) are released due to corrosion. The absorption of mercuric ions from gastrointestinal tract is minute, therefore, this form has very little toxic effect. However, mercury ions play a role in allergic reactions.

Mercury vapors are carried to lungs from intraoral air, where it is rapidly absorbed into bloodstream. This form is rapidly oxidized from body.

Mercury exposure in dental office can result from:

- Incorrect storage of mercury or waste amalgam
- Spillage of mercury
- Mixed, but unhardened dental amalgam
- During finishing and polishing of amalgam
- Removal of amalgam

Monitoring Mercury Levels

Several methods, which can determine concentration of mercury are:

- Mercury vapor may be determined by using a mercury detection meter, such as mercury sniffer.
- Paper discs impregnated with palladium chloride can be used. (The disadvantage is lack of reaction specificity for mercury.)
- A badge system may be used in which mercury is adsorbed on gold foil.
- Mercury in vapor and dust form may be determined by passing the known volume of air through an absorbing system and then quantifying the absorbed mercury.

Mercury Allergy

Allergy to mercury is very rare. Mercury hypersensitivity is an allergic response mediated by immune system. It is a cell-mediated response. Delayed hypersensitivity to mercury results in contact eczematous reactions on skin and mucosa. Mercury allergy can be confirmed by patch test.

Amalgam restorations may also induce allergic reactions. Women are more affected than men. Symptoms include dermatitis, urticaria, eczema, erythema, edema and itching occurring in face, neck and limbs. Oral manifestations include gingivo-stomatitis, hypersalivation and erosive lesion.

Mercury Toxicity

Mercury from amalgam fillings have been evaluated for its toxicity related to nervous tissue, kidney, immune system and general health of patient; however, no cause and effect relationship have been found.

Mercury Hygiene

- Mercury-containing products should not be stored in open, but in cabinets. Storage locations should be near a vent that exhausts air out of building.
- Local spill or spatters of triturated materials should not be collected with a vacuum aspiration. Spilled mercury can be made harmless by dusting with sulfur powder or spraying with a solution of sodium thiosulfate.
- Mercury droplets are very difficult to remove from artificial floor coverings, so such coverings should either be avoided or intermittently replaced.
- To control the vapors of mercury during placement and condensation procedures, rubber dam should be used to isolate the patient.
- Scrap dental amalgam from condensation procedures should be collected and stored under water, glycerine or spent X-ray fixer in a tightly capped jar.
- Silver mercury has a very low melting point and easily melts during finishing. (Amalgam should be polished at slow speed using water spray.)
- Instruments used for inserting, finishing, polishing or removing dental amalgam may contain some amalgam material on their surface. During instruments sterilization; this material may be heated and release mercury vapor.
- Provide proper ventilation in work place by having fresh air exchanger and proper replacement of filters, which may act as trap of mercury.
- Mercury vapor level in office should be periodically monitored.
- Office personnel should also be periodically monitored.
- Pre-capsulated alloys should be used to eliminate the possibility of bulk mercury spill.
- Amalgamators should be covered.
- Skin contact with mercury or freshly prepared amalgam should be avoided.
- Change masks after removing amalgam restorations.
- Mercury contaminated items should be deposited in sealed bags.
- Spilled mercury should be cleaned with trap bottles/tapes.
- In dental operatory, professional clothing should be worn.

The evidences do not demonstrate that amalgam restorations are a hazard to health; however, occupational hazard to mercury is a potential hazard for dental personnel. The operator should observe strict mercury and amalgam hygiene procedures.

BIBLIOGRAPHY

1. Aminzadeh KK, Etminan M. Dental amalgam and multiple sclerosis: A systematic review and meta-analysis. J Public Health Dentistry 2007;67.
2. Berglund A, Pohl S, Olsson S, Bergman M. Determination of the rate of release of intraoral mercury vapour from amalgam. J Dent Res 1988;67:1235–39.
3. Blair FM, JM, McCabe JF. The physical properties of a gallium alloy restorative material. Dent Mater 1995; 11:277–82.
4. Browning WD, Johnson WW, Gregory PN. Clinical performance of bonded amalgam restorations at 42 months. JADA 2000;131:607–10.

5. Chin G, Chong J, Kluczewska AL, Gorjy S, Tennant M. The environmental effects of dental amalgam. Aust Dent J 2000;45:246–50.
6. Dunne SM, Wilson NHF, Gainsford ID. Current materials and techniques for direct restoration in posterior teeth. Part I-silver amalgam Int Dent J 1997;47:123–28.
7. Holmstrup P. Oral mucosa and skin reactions related to amalgam. Adv Dent Res 1992;6:120–23.
8. Jordon RE, Suzuki M, Boksman L. The new generation amalgam alloys: clinical considerations. Dent Cl North Am 1985;29:341–46.
9. Knight GT, Berry TG. Clinical application of a direct placement mercury-free alloy. Am J Dent 1997;10:52–56.
10. Mahler DB, Adey JD, Marek M. Creep and corrosion of amalgam. J Dent Res 1982;61:33–36.
11. Mamoun JS, Ahmed M. Amalgam matrix for class II and class V preparations connected at the proximal box. JADA 2006;137:186–89.
12. Navarro MFL, Franco EB, Bastos PAM. Clinical evaluation of gallium alloy as a posterior restorative material. Quint Int 1996;27:315–19.
13. Oddera M. Conservative amalgam restoration of class II lesions-The 'slot restoration' A case report. Quint Int 1994; 25:493–96.
14. Papa J, Cain C, Masser HH, Wilson PR. Tunnel restorations versus class II restorations for small approximal lesions: a comparison of tooth strengths. Quint Int 1993;24:93–96.
15. Rasheed AA. Effect of bonding amalgam on the reinforcement of teeth. J Prosth Dent 2005;93:51–55.
16. Roberts HW, Charlton DG, Murchison DF. Repair of non carious amalgam margin defects. Oper Dent 2001;26:273–77.
17. Shaini FJ, Fleming GJP, Shortall ACC, Marquis PM. A comparison of the mechanical properties of a gallium-based alloy with a spherical high copper amalgam. Dent Mater 2001;17:142–46.
18. Spencer AJ. Dental amalgam and mercury in dentistry. Aust Dent J 2000;45:224–27.
19. Summitt JB, Osborne JW, Burgess JO, Howell ML. Effect of grooves on resistance form of class II amalgams with wide occlusal preparations. Oper Dent 1993;18:42–46.
20. Williams P. Goodbye Amalgam, Hello Alternatives? Dent Mater 1996;62:139–41.
21. Zyman P. Adhesive amalgam restoration technique: Clinical tips. J Esthet Dent 1997;9:131–34.
22. Martin J, Eduardo F, Estay J, Gordon V, Mjor IA, Moncado G. Management of class I and class II amalgam restorations with localized defects: Five-year results. Int J Dent 2013 (Article ID 450260).
23. Roggenkamp CL, Berry FA, Lu H. In vitro bond strengths of amalgam added to existing amalgams. Oper Dent 2010; 35:314–23.
24. Zguri MN, Casey JA, Jessup JP, Vandewalle KS. Proximal contact repair of complex amalgam restorations. Oper Dent 2017;42:266–72.